

JIAJIE JIAN

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EDUCATION

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- **Chongqing University** Chongqing, China
Master of Information and Communication Engineering Sep. 2024 – Expected Jun. 2027
 - **Co-advisors:** Prof. Shu Fu & Prof. Min Liu
 - **Grade:** 88.57/100
 - **Core Courses:** Modern Signal Processing, Information Theory and Coding, Deep Learning, Radar Systems and Signal Processing
 - **Sichuan Normal University** Chengdu, China
Bachelor of Electronic Information Engineering Sep. 2020 – Jun. 2024
 - **Grade:** 84.16/100
 - **Core Courses:** Signals and Systems, Digital Signal Processing, Principles of Communications, Probability and Statistics, Linear Algebra

RESEARCH INTERESTS

Integrated Sensing and Communications (ISAC); Physical Layer Security; AI-Enabled Wireless System Design; Optimization for Wireless Communications; Statistical Signal Processing.

PUBLICATIONS AND MANUSCRIPTS

Journal Publication

- [J1] **Jiajie Jian**, Min Liu, Shu Fu, and Yongming Huang, “Bidirectional Covertness and Security Enhancement in FD-ISAC Systems,” *IEEE Transactions on Vehicular Technology*, 2026. [\[DOI\]](#)

Manuscripts Under Review / Submitted

- [M1] **Jiajie Jian**, Zan Li, Shu Fu, Jiangbo Si, and Min Liu, “Cross-Boundary Covert UAV Service Near No-Fly Zones: Risk-Aware Trajectory and Scheduling,” *IEEE Transactions on Mobile Computing*, revision under review.
- [M2] **Jiajie Jian**, Shu Fu, Tingting Chen, and Min Liu, “Resilient Beamforming for Anti-Jamming MIMO-ISAC,” *IEEE Wireless Communications Letters*, submitted.

RESEARCH EXPERIENCE

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- **Sensing-Guided Beamforming for MIMO-ISAC under Unknown Jamming [M2]** Feb. 2026 – Present
 - Developed a sensing-guided beamforming framework to mitigate communication and tracking degradation in MIMO-ISAC under unknown jamming, using sensing observations to construct a spatial interference surrogate for proactive defense.
 - Reduced the joint communication-sensing beam update to a search over two dual variables by exploiting its Kronecker structure, cutting dimensionality by **>95%** and runtime by **>86%** versus a direct CVX solver implementation.
 - Built a lightweight learning-based selector that adapts defense parameters online from sensing feedback, improving effective sum rate by **18%** over fixed robust designs under reactive attacks.
 - Benchmarked against representative baselines and performed ablations under varying attack intensities, identifying null broadening and dedicated sensing DoFs as key mechanisms for mitigating unknown jamming.

- **Risk-Aware UAV Trajectory and Scheduling for Wireless Service [M1]** Sep. 2025 – Present
 - Designed a joint trajectory-and-scheduling framework for UAV service under constrained mobility, balancing reliable connectivity with mission-wide control of unintended RF leakage near regulated areas.
 - Developed a mission-level RF-leakage risk model beyond average exposure metrics, converting slot-level exposure events into an additive budget that links spatial, temporal, and transmission decisions.
 - Used the budget’s additive structure to derive closed-form per-slot scheduling updates via Lagrangian relaxation, then jointly updated trajectory, rate, and leakage variables through convex approximation.
 - Evaluated the joint design through key-parameter sweeps and 2-D visualizations of optimized UAV trajectories and user scheduling, achieving **>40%** total spectral efficiency gains over heuristic scheduling and about **15%** over an optimized fixed-trajectory baseline.
- **Diffusion-Assisted EKF Tracking for UAV-ISAC Beam Alignment** Mar. 2026 – Jun. 2026
 - Augmented an EKF tracker with a diffusion model for agile UAV maneuvers, using historical states to forecast nonlinear motion and reduce tracking drift.
 - Used normalized innovation squared (NIS) as an online control signal to adapt DDIM denoising depth to model mismatch, suppressing drift without unnecessary inference.
 - Built the end-to-end PyTorch workflow—from model training and DDIM inference to EKF integration—and delivered the upstream state-estimation module for a collaborative UAV-ISAC beamforming study.
 - Validated the tracker on 30 UAV trajectories, reducing position, azimuth, and elevation RMSE by at least **54%** in forecasting and **31%** in continuous tracking relative to the underlying EKF tracker.
- **Joint Transceiver Design for Secure Full-Duplex ISAC [J1]** Dec. 2024 – Aug. 2025
 - Developed a joint FD-ISAC transceiver design for bidirectional security, mitigating radar-induced downlink exposure and balancing public traffic’s masking benefit against its interference penalty for uplink protection.
 - Used detection monotonicity to separate surveillance-channel uncertainty from beamforming–power coupling, converting robust constraints into linear matrix inequalities via the S-procedure and approximating the remaining nonconvex terms with tight convex surrogates.
 - Derived a zero-forcing implementation that separated beam directions from power allocation, reducing large-array design complexity from $\mathcal{O}(N^{6.5})$ matrix optimization to an $\mathcal{O}(N^2)$ search over two power variables.
 - Designed and conducted simulations across security, sensing, and channel-uncertainty regimes, achieving bidirectional protection through native-signal reuse and embedded artificial noise, with no dedicated security hardware.

LEADERSHIP & MENTORSHIP

- **Undergraduate Research Mentorship** Oct. 2025 – Apr. 2026
 - Mentored an undergraduate in DRL-based trajectory optimization for exposure-aware UAV communications, providing end-to-end guidance from literature review to simulation design.

SKILLS

- **Methods:** Optimization Theory, Statistical Signal Processing, Matrix Analysis, Machine Learning
- **Programming & Tools:** Python (PyTorch), MATLAB (CVX), LaTeX
- **Languages:** English (IELTS: 7.5 / CEFR C1), Mandarin (Native)

HONORS & AWARDS

- First-Class Academic Scholarship, Chongqing University 2024, 2025
- First and Second-Class Academic Scholarship, Sichuan Normal University 2021, 2023